The Migration of Extremes

Tracking Compound Drought-Heatwaves in 3D Space

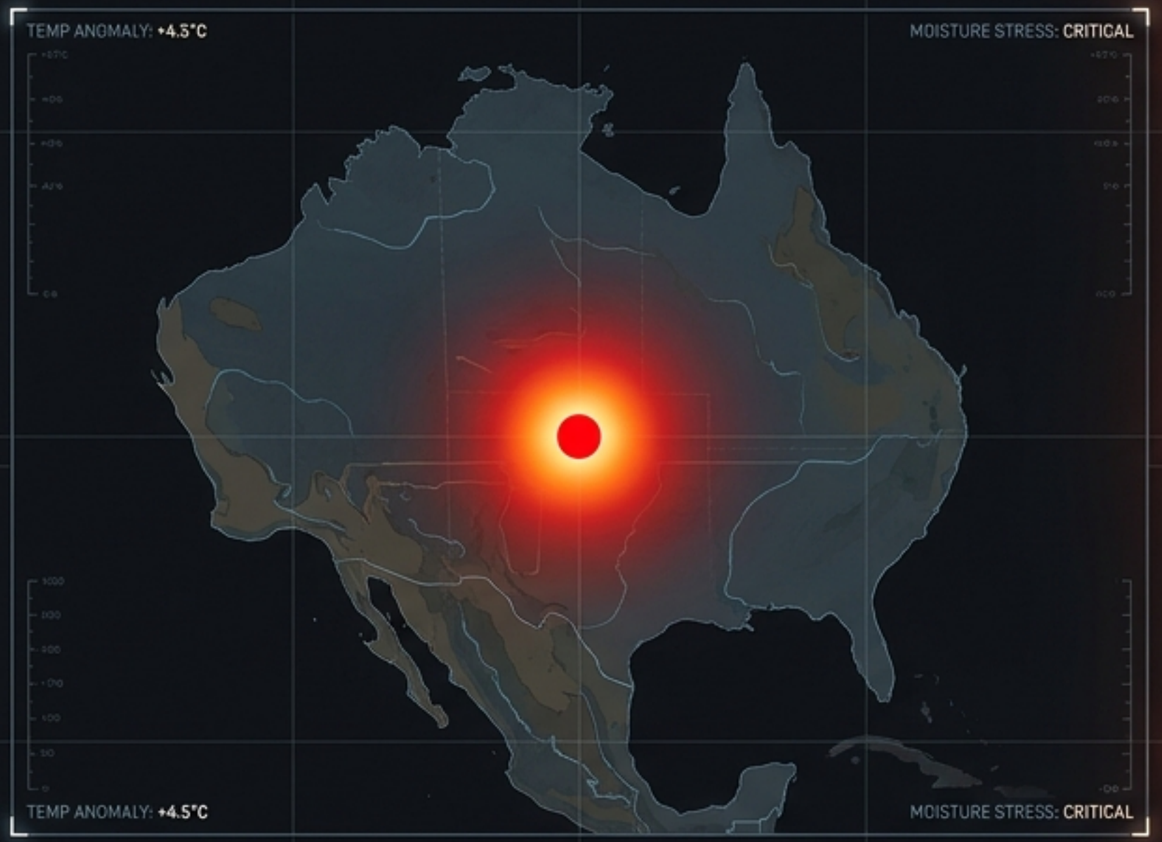
An analysis of 2,584 contiguous climate events (1982–2019)

From Static Observation to Dynamic Anticipation



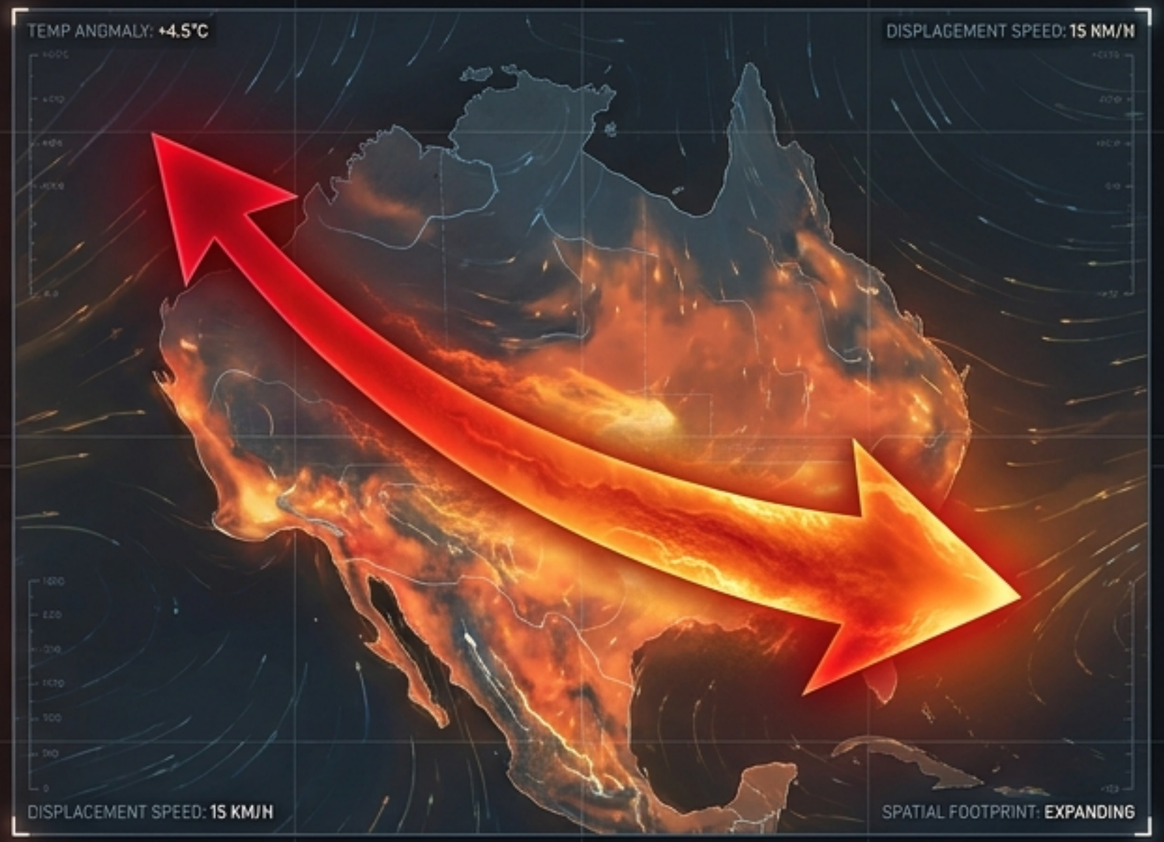
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THE STATIONARY THREAT



Historically, compound drought and heatwaves (CDHWs) have been treated as spatially stationary extremes—places where a region gets “stuck” in bad weather.

THE MIGRATING PREDATOR

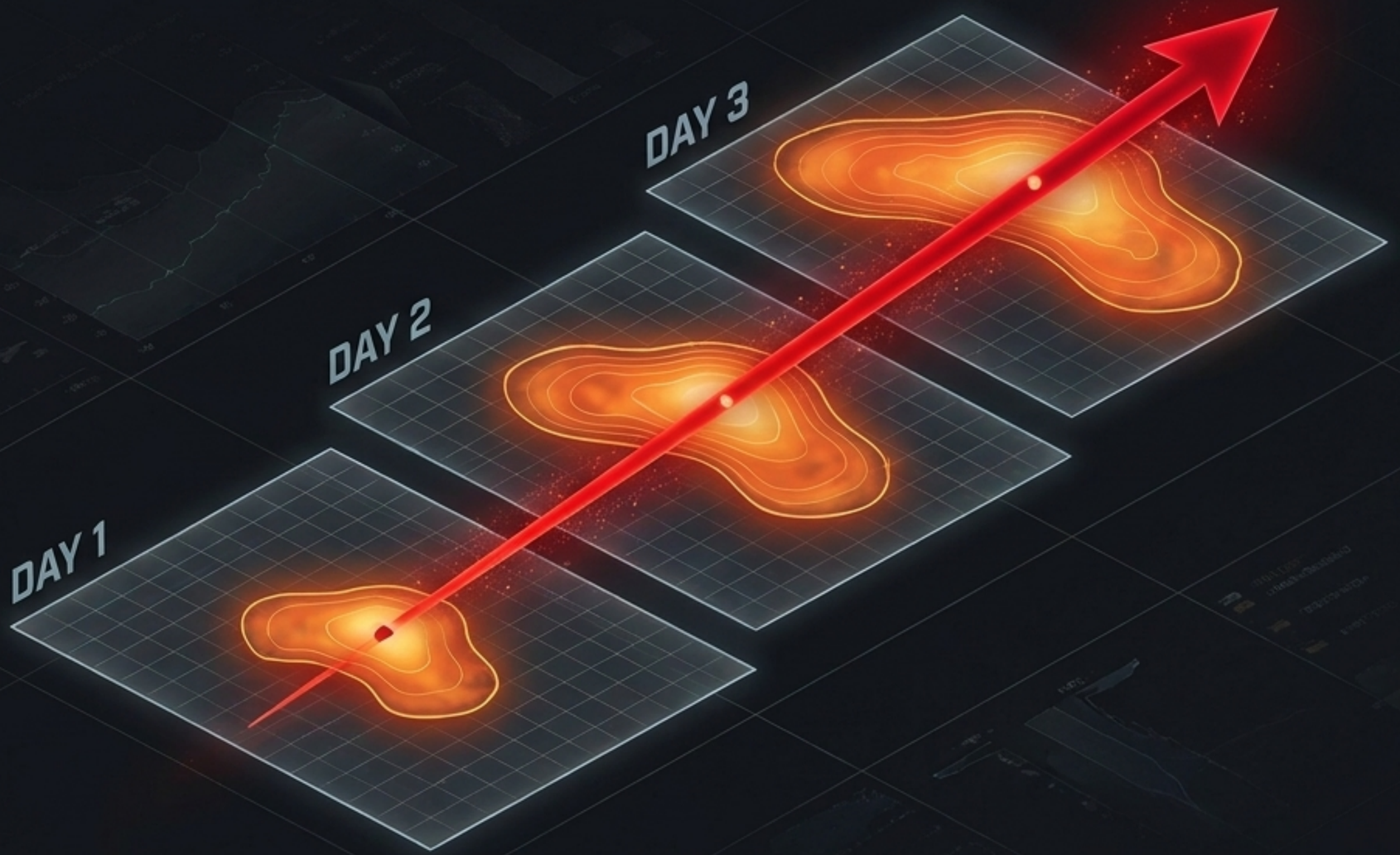


New modeling reveals CDHWs exhibit systematic displacement. They are massive, self-sustaining entities that travel hundreds of kilometers, expanding their footprint of compound heat and moisture stress.

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THE EASIEST WAY TO TRACK A CDHW

Connecting the dots in 3D space across time and geography.



THE FILTER

Tracking requires a footprint **>1 million km²** lasting **>3 consecutive days**.

THE RESULT

2,584 distinct, contiguous events isolated globally between 1982 and 2019.

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THE GLOBAL HIGHWAYS OF HEAT

Migration is not random; it follows the atmospheric jet streams.



NORTHERN HEMISPHERE DOMINANCE

Events here expand up to ~7×10^6 km², driven by transient weather systems and stronger temperature variability.

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THE RULES OF MOVEMENT

The geometry of a CDHW dictates its future behavior.



Rule 1: **Time = Distance**

(Longer events travel further)

In North America, for every additional day of event duration, the migration distance increases by approximately 127 km. (S. America shows the strongest correlation at $R^2 = 0.87$).



Rule 2: **Size = Speed**

(Larger events travel faster)

Massive spatial extents generate their own atmospheric momentum. In Africa, every 100,000 km² increase in maximum area accelerates the event by ~10 km/day.

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DIAGNOSING THE ENGINE

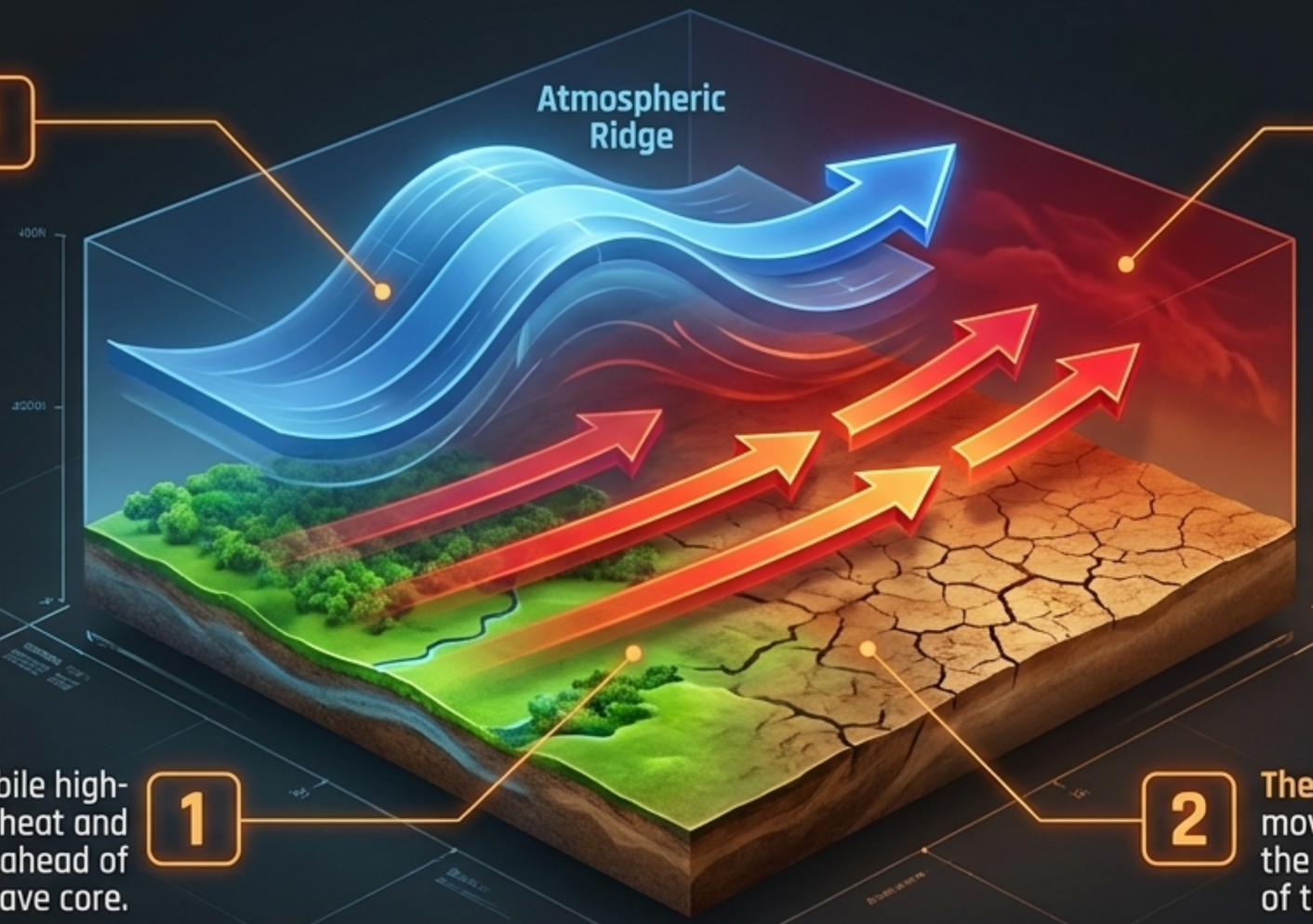
Distinguishing between moving threats and stationary domes.

	THE MIGRATORY EVENT	THE LOCALIZED EVENT
ATMOSPHERIC DRIVER	Mobile Rossby Waves (Upper-level ridges shifting downstream).	Quasi-Stationary “Omega Blocks” (Trapped high-pressure systems).
MEDIAN DISPLACEMENT	~800 km (High mobility).	~500 km (Confined movement).
PEAK INTENSITY (SEVERITY)	Moderate (~5.0).	Severe (~7.5) due to accumulating in situ feedback.
PRIMARY IMPACT MECHANISM	Pushes heat forward, pre-drying land ahead of the core.	In-place amplification of sensible heat and soil desiccation.

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ANATOMY OF MIGRATION: THE “SNOWPLOW” EFFECT

How a mobile atmospheric ridge paves its own path of destruction.



1

The Push: A mobile high-pressure system forces heat and dry air downstream (+0.093 K/s heat advection).

3

The Migration: The heatwave effortlessly follows the newly created dry corridor, perpetually feeding on the depleted landscape.

1

The Push: A mobile high-pressure system forces heat and the atmosphere and soil ahead of the actual heatwave core.

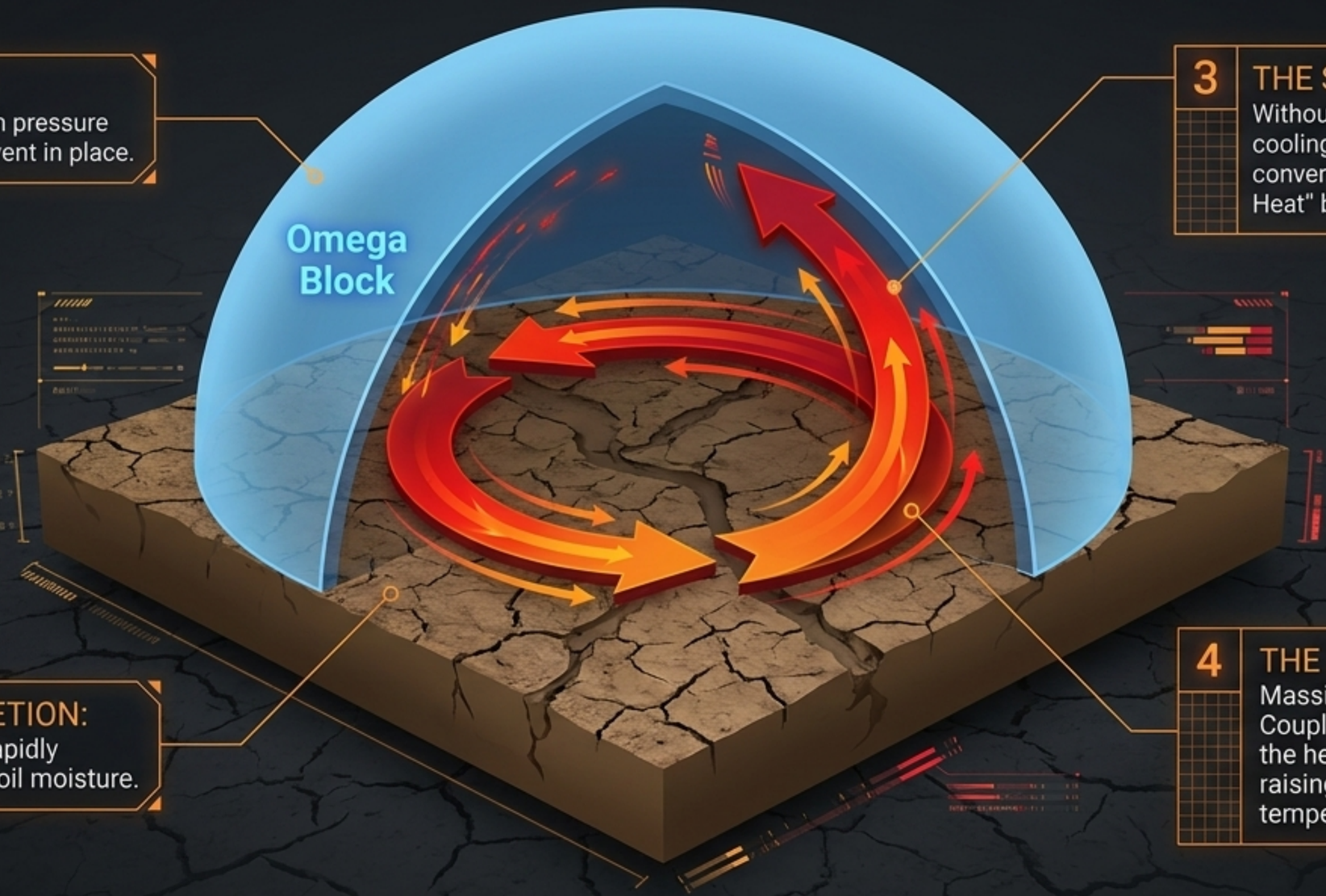
2

The Pre-Drying: This movement strips moisture out of the atmosphere and soil ahead of the actual heatwave core.

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ANATOMY OF THE HEAT DOME

The self-reinforcing feedback loop of a stationary event.



1 THE TRAP:
Stationary high pressure anchors the event in place.

2 THE DEPLETION:
Intense sun rapidly vanishes all soil moisture.

3 THE SHIFT:
Without moisture, evaporative cooling halts. The energy converts entirely into "Sensible Heat" blasting upward.

4 THE AMPLIFICATION:
Massive Land-Atmosphere Coupling (+2.72) forces the heat back back down, raising surface temperatures exponentially.

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THE ANCHOR PARADOX

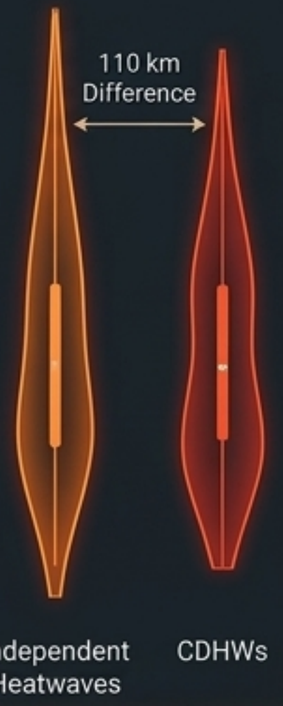
How drought physically restricts the movement of extreme heat.



HUMID REGIONS:

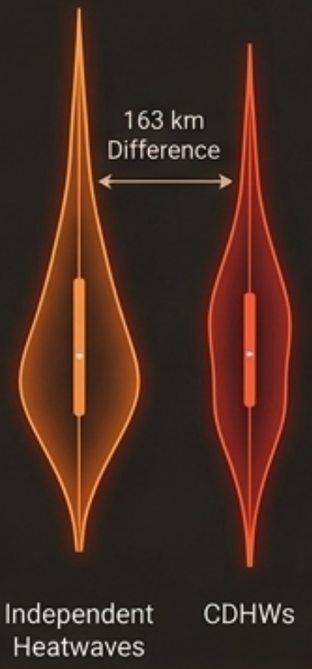
Minimal anchoring (Difference: 110 km).

Plentiful moisture allows the heatwave to maintain momentum.



ARID REGIONS:

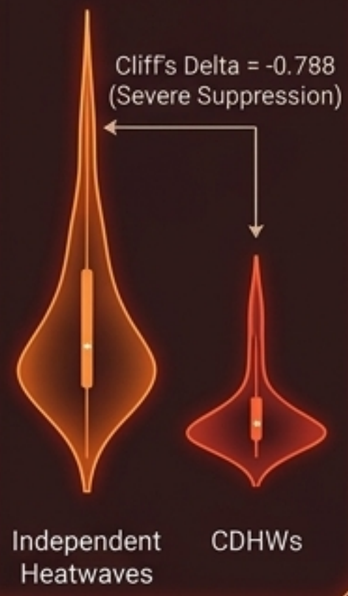
Moderate anchoring (Difference: 163 km).



SEMI-ARID REGIONS:

Severe anchoring (Cliff's Delta = -0.788).

Heatwaves are highly suppressed here; the rapid loss of marginal soil moisture creates intense local heating that kills atmospheric instability, trapping the event in place.

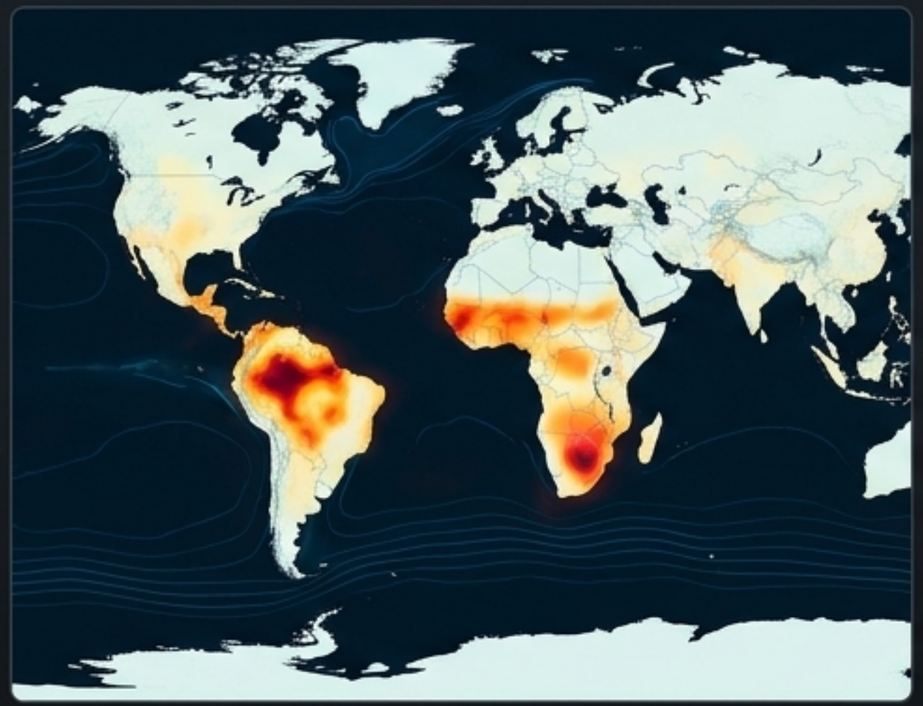


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The Expanding Danger Zones

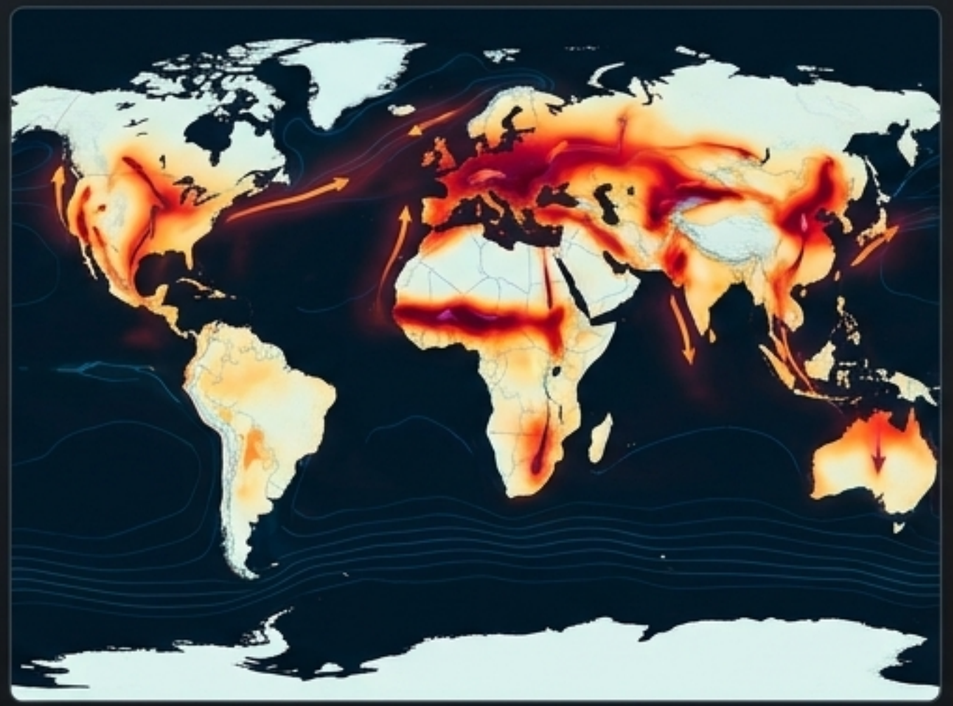
Migration corridors are widening, aggressively targeting new latitudes.

1982–2000



Migration pathways are largely confined to the Amazon basin, Western Africa, and Southern Africa.

2001–2019



An explosion of deep crimson migration tracks across the mid-latitudes.

The Emerging Hotspots

- The European Plains (Massive intensification)
- Western North America
- The West African Sahel
- Central Asia & Northern China
- Northern Australia

The Driver: Increased anthropogenic warming and amplified atmospheric moisture demand have fundamentally altered global migration corridors.

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Assessing Land Vulnerability

Why certain landscapes act as corridors for migrating extremes

Croplands



Amplification Factor: Low to Moderate (< 2x in most regions, peaking at 4.5x only in highly dense 80% coverage).

Absolute Impact: Modest (+3 additional migration tracks).

The Physics: Deeper root systems and access to irrigation provide a minor buffering capacity against rapid evaporative shock.

Pasturelands



Amplification Factor: Severe and Consistent (**> 2x everywhere**, gaining **5 to 7+** tracks in moderate 40–50% coverage).

The Physics: Shallow grass roots rely entirely on near-surface moisture. They undergo rapid desiccation, instantly killing local evaporative cooling.

The Result: Pasturelands act as highly sensitive, dry "runways" that actually accelerate CDHW migration.

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The Tipping Point

The permanent transition from pastureland to scrubland.



Data Story

The Red Line (Land-Atmosphere Coupling): Shows a steady climb from a baseline of ~0.65 (pre-2000) to ~0.75 (post-2020). The land is literally baking the air above it with greater intensity.

The Blue Bars (Scrubland Conversion): A massive spike. Prior to 2000, permanent degradation affected <0.5% of global pastureland. By the 2020s, it surged past 1.5%.

Insight Box

Repeated CDHW migration tracks are not just temporary weather events; they are fundamentally destroying ecosystem resilience, permanently transitioning grazing lands into arid scrub.

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The Macro Domino Effect

How migrating extremes engineer their own future pathways.



1. Global Warming Shifts Corridors

Increased background heat alters atmospheric circulation, expanding CDHW migration pathways.

2. Events Target Pasturelands

Migrating CDHWs follow the path of least resistance across vulnerable, shallow-rooted grazing lands.

3. Land Undergoes Permanent Degradation

Repeated hits exhaust the ecosystem, triggering the pasture-to-scrubland tipping point.

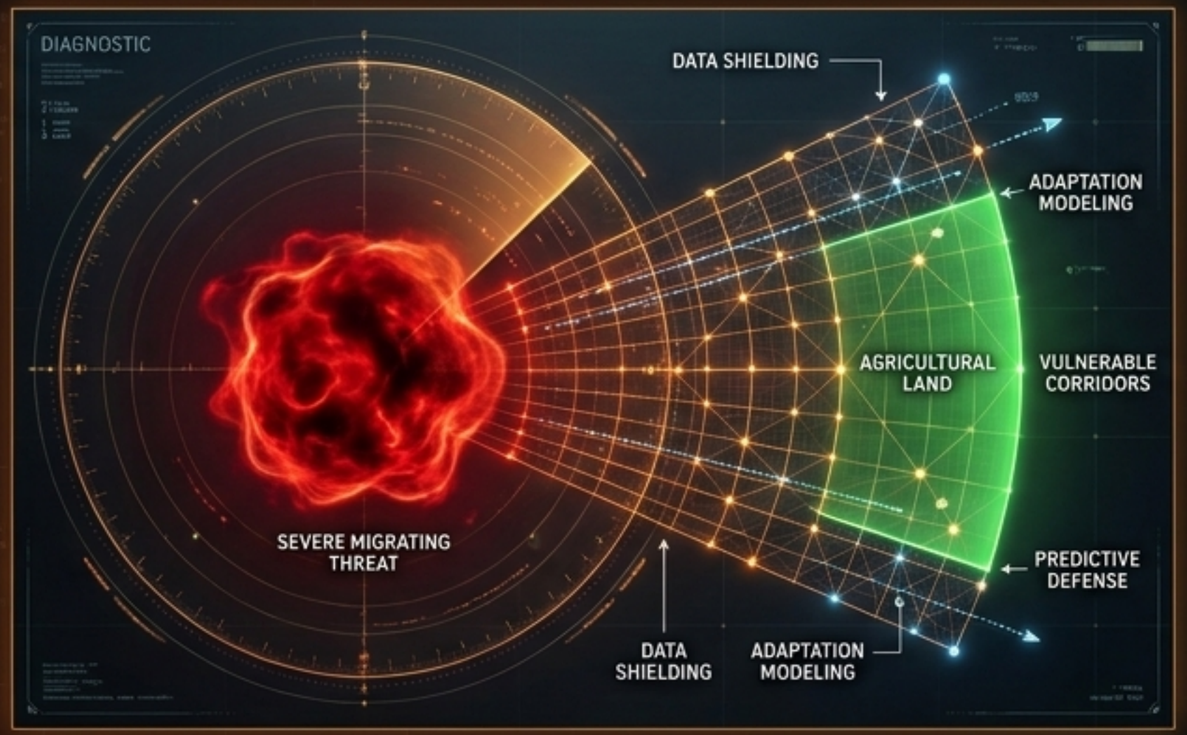
4. Degraded Land Fuels Future Extremes

The newly created scrubland loses its moisture-buffering capacity, acting as an anchor that fuels future, localized "Heat Domes".

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From Observation to Anticipation

The future of global resilience requires trajectory-aware defense.



Look Upwind, Not Just Up: We can no longer prepare by looking solely at local forecasts. We must track high-pressure systems hundreds of miles away.

Trajectory-Aware Early Warning: Anticipating downstream impacts before local onset is the only way to safeguard vulnerable agricultural corridors.

Targeted Land-Use Policy: Strategic agricultural planning must account for these newly established “migration tracks” to break the cycle of degradation.

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